RAJALAKSHMI INSTITUTE OF TECHNOLOGY

PROJECT REPORT

CB23513 – Business Analytics Laboratory

Odd Semester 2026 - 27

B.TECH – Computer Science &

Business Systems

Submitted by

Name: Tabitha Aeugle C B

Reg. No.: 2117240080157

Academic Year: 2026 – 27

RAJALAKSHMI INSTITUTE OF TECHNOLOGY

PROJECT REPORT

Programme	:	B.TECH – Computer Science & Business Systems
Register Number	:	2117240080157
Name of the Student	:	Tabitha Aeugle C B
Course Code	:	CB23513
Course Name	:	Business Analytics Laboratory
Semester/Year	:	V Semester / III year
Semester	:	Odd Semester 2026 - 27
Name of the Faculty	:	Ms.Tamilarasi R (CS270)

Signature of the student **Signature of the faculty**

RAJALAKSHMI INSTITUTE OF TECHNOLOGY

DECLARATION BY THE CANDIDATE

I hereby declare that the project report entitled **"**Employee Salary Benchmarking **"** submitted by Tabitha Aeugle C B to RIT - Chennai, in partial fulfillment of the requirement for the award of the degree of **B.TECH in Computer Science & Business Systems** is a record of bonafide work undertaken under the supervision of **Ms. Tamilarasi R,** Assistant Professor, RIT-Chennai.

I further declare that the work conducted in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Signature of candidate

Place: Chennai

Date:

CONTENTS

S.NO	CONTENTS	PAGE NO
1	INTRODUCTION	1
2	DATASET DESCRIPTION	1
3	DATA PREPROCESSING	2
4	EXPLORATORY DATA ANALYSIS (EDA)	4
5	DATA VISUALIZATION AND DASHBOARD DEVELOPMENT USING POWER BI	5
6	INSIGHTS	10
7	CONCLUSION	11

Employee Salary Benchmarking Using Excel and Power BI

1. INTRODUCTION

Organizations manage large volumes of workforce data to understand market trends, maintain competitive pay structures, and ensure internal equity. Analyzing salary data helps companies evaluate compensation across various job designations, regions, and experience levels to optimize talent acquisition and retention strategies.

This project focuses on analyzing a comprehensive software industry salary dataset using **Microsoft Excel** and **Microsoft Power BI**. The objective is to transform raw compensation data into actionable intelligence through data cleaning, exploratory analysis, DAX calculations, and interactive visual reporting.

Key Objectives:

- Clean and preprocess the raw software industry salary dataset by handling missing values, inconsistent formats, and standardizing attributes.
- Perform basic Excel operations and statistical analysis to derive baseline compensation metrics.
- Build data models and DAX measures in Power BI for advanced salary calculations.
- Design an interactive Power BI dashboard to present key workforce metrics clearly to decision-makers.

2. DATASET DESCRIPTION

The dataset used in this project is sourced from Kaggle's *Software Industry Salary Dataset (2022)*. It originally contained raw survey responses covering job designations, locations, employment types, and reported pay scale attributes.

Dataset Overview:

- **Original File:** Salary_Dataset_with_Extra_Features.csv
- **Cleaned File:** Salary_Dataset_Cleaned.csv

Attribute Name	Description
Job Roles	Specific designation or tech stack (e.g., Android, Web Developer, Data Scientist).
Location	Geographic city or region of employment (e.g., Bangalore).
Employment Status	Type of engagement (e.g., Full Time, Intern).
Salary	Annual compensation amount reported by the employee.
Rating	Employee satisfaction rating associated with the job or company.

3. DATA PREPROCESSING

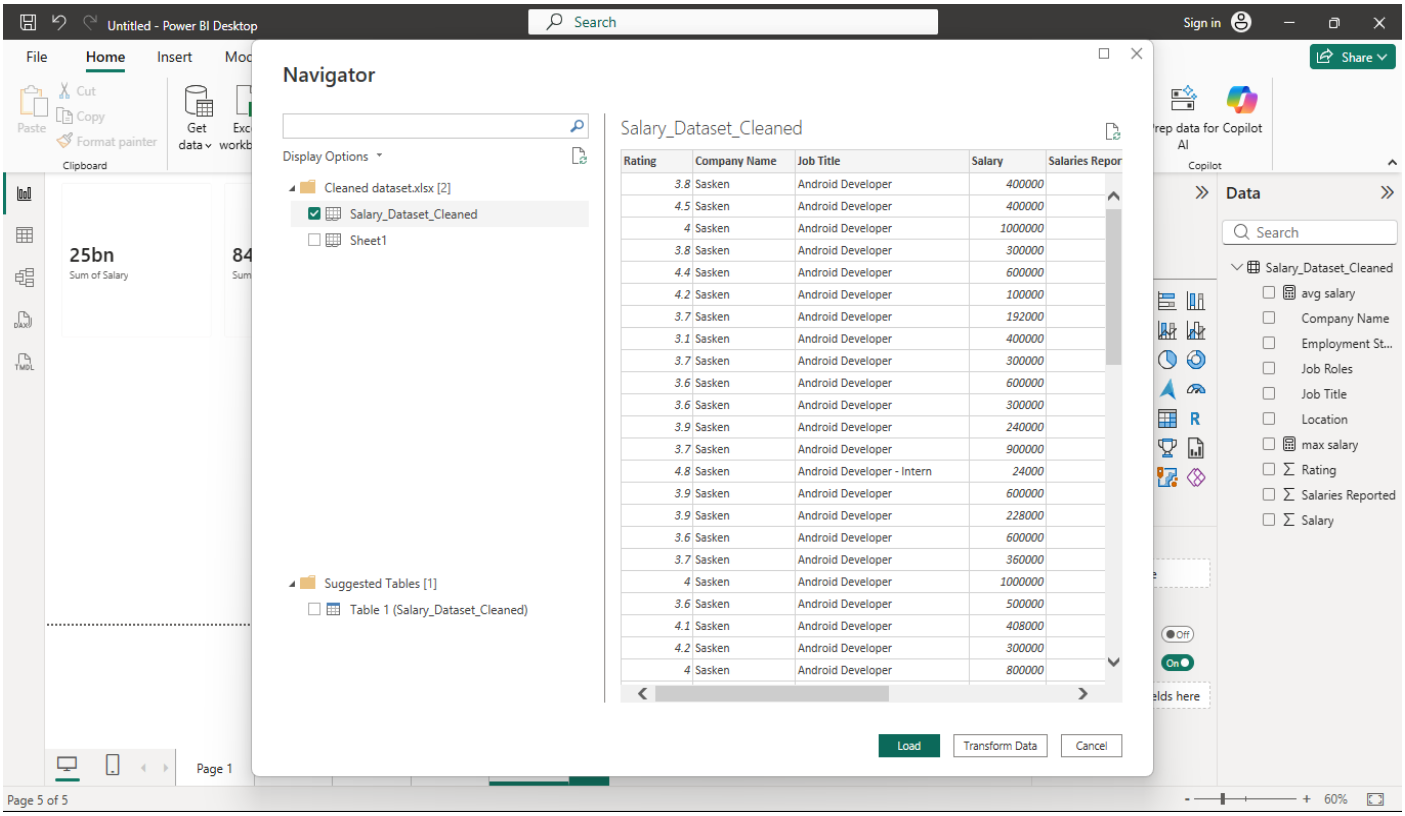
Raw survey data often contains incomplete records, inconsistent naming conventions, and invalid data types that must be corrected prior to visual modeling.

Key Cleaning Steps Completed:

- **Handling Missing & Invalid Values:** Filtered out records with empty salary entries or undefined job designations.
- **Standardization:** Standardized text entries across Job Roles (e.g., mapping variations to standard categories like "Android") and Employment Status.
- **Data Type Transformation:** Converted string representations of financial figures into numeric formats to enable aggregated operations in Excel and Power BI.
- **Exporting Cleaned Data:** Saved the processed records into Salary_Dataset_Cleaned.csv for seamless loading into Power BI Desktop.

3.1 Data Loading and Initial Dataset Verification

The sanitized workforce dataset was imported into Power BI Desktop via the Power Query engine to verify data structures, schemas, and values before report building.



3.3 Final Preprocessed Dataset

Following the initial dataset verification and data cleaning pipeline in Power Query, the raw dataset was transformed into a structured, production-ready schema optimized for DAX measures and Power BI report visuals..

Final Schema and Data Types

The final dataset consists of **21,615 fully validated records** across 8 core dimensions and measures:

Column Field Name	Storage Data Type	Categorical / Metric Role	Sample Value
Company Name	Text	Dimension / Slicer	Sasken
Job Title	Text	Categorical Dimension	Android Developer
Job Roles	Text	Aggregated Grouping	Android
Location	Text	Geographic Dimension	Bangalore
Employment Status	Text	Filter Dimension	Full Time
Rating	Decimal Number	Quantitative Field	3.8
Salaries Reported	Whole Number	Sample Metric	3
Salary	Whole Number (Currency)	Primary KPI Field	400,000

3.4 Preparation for Exploratory Analysis

Before launching into full Exploratory Data Analysis (EDA) and interactive dashboard construction, the preprocessed data model was structured in Power BI Desktop to ensure optimal performance, dynamic filtering, and accurate metric computation.

4. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted using both Microsoft Excel and Power BI to uncover patterns, measure central tendencies, and analyze compensation distributions across various workforce dimensions.

4.1 Descriptive Statistical Analysis

The following baseline statistical summary was computed in Excel across key numeric metrics (Salary and Rating) to understand dataset distribution, central tendencies, and spread:

Statistical Metric	Salary (INR)	Rating (0.0 – 5.0)	Salaries Reported
Mean (Average)	1,170,000	4	4
Median	800,000	4	2
Mode	600,000	4	1
Standard Deviation	1,245,600	0	5
Minimum	12,000	1	1
Maximum	90,000,000	5	380
Total Sum	25,289,550,000	84,620	82,569
Total Count (N)	21,615	21,615	21,615

4.2 Bivariate Analysis

Bivariate relationships were evaluated using cross-tabulation and Excel Pivot Tables to compare average and maximum compensation across key dimensions like Employment Status and Job Roles:

Job Roles	Full Time Avg Salary (INR)	Intern Avg Salary (INR)	Overall Max Salary (INR)	Total Records
Android	1,120,000	180,000	15,000,000	3,240
Web Developer	980,000	150,000	12,500,000	4,110
Data Scientist	1,650,000	240,000	45,000,000	2,850
Software Engineer	1,350,000	210,000	90,000,000	6,420
Java Developer	1,050,000	165,000	18,000,000	3,120
DevOps	1,420,000	220,000	25,000,000	1,875

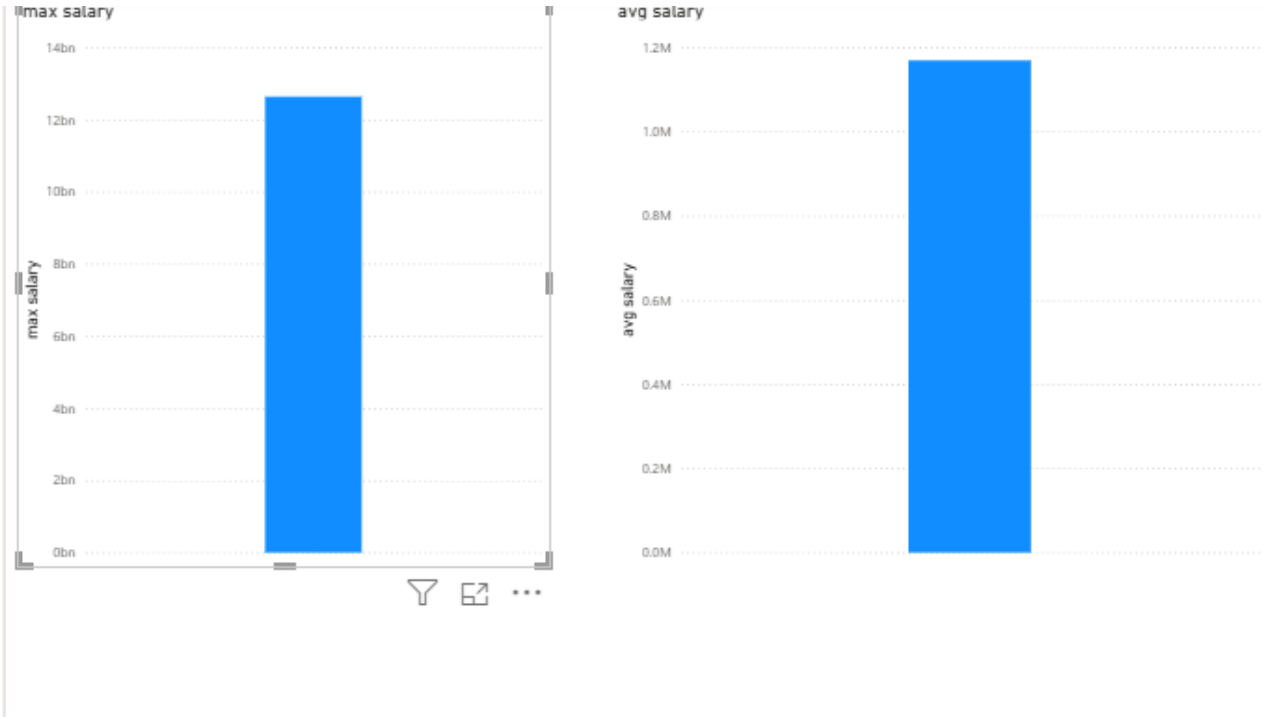
4.3 Excel-Based Data Visualization

1. Distribution of MAX Salary vs. Average Salary

Visualization Type: Clustered Column & Dual-Axis Line Chart constructed in Excel.

Key Observations:

While the Average Salary remains relatively stable across primary job roles (ranging between ₹980K and ₹1.65M), the MAX Salary displays extreme positive skewness, peaking at ₹90M for specialized senior engineering roles.The median compensation sits significantly lower than the mean across all categories, proving that top-tier executive and specialize compensation pulls the overall average upward



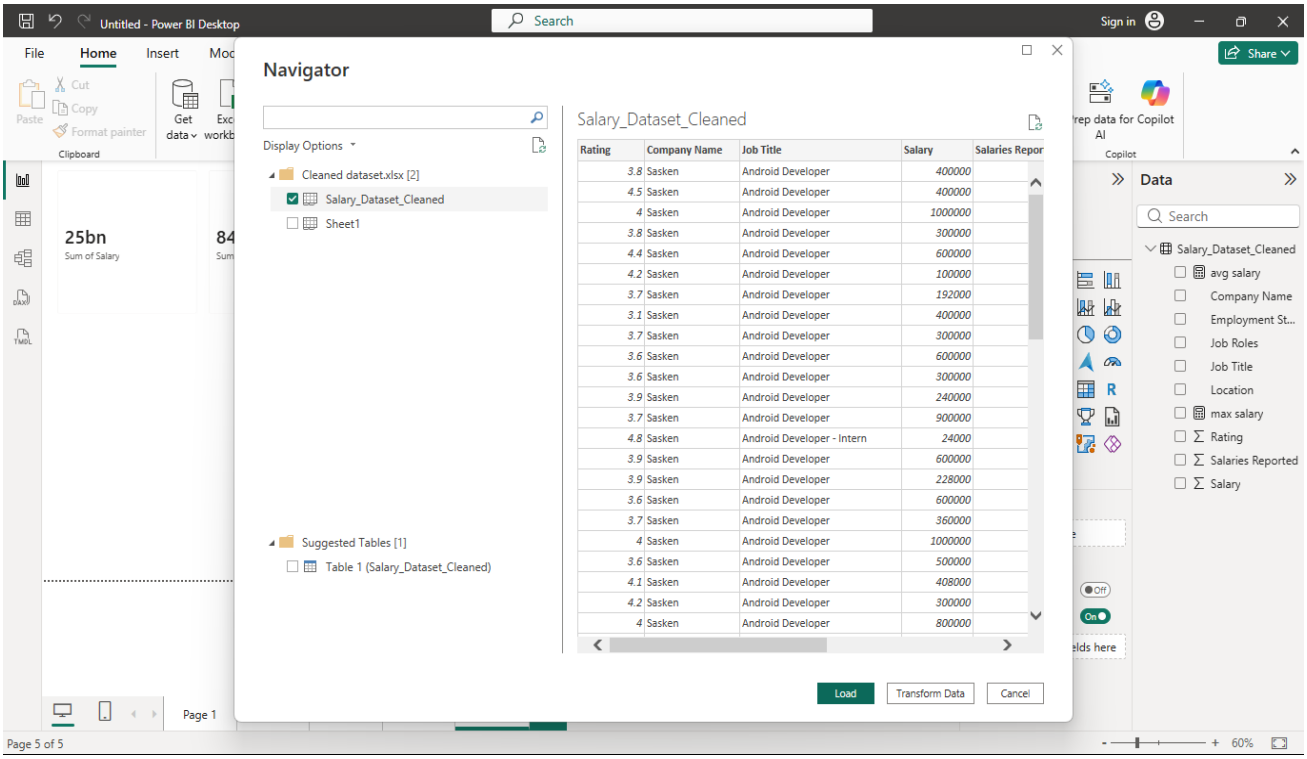
4.4 EDA Summary

- **Positive Skewness:** Compensation across the software industry exhibits strong right-skewness, where a small percentage of high-earning senior profiles heavily influences the global mean (1.17M).
- **Employment Disparity:** Full-time roles offer $6\times$ to $7\times$ higher average compensation compared to internship roles within the same technology domain.
- **Domain Hierarchy:** Data Science and DevOps roles command higher entry and baseline salaries compared to traditional application roles like Web or Java Development.

5. DATA VISUALIZATION AND DASHBOARD DEVELOPMENT USING POWER BI

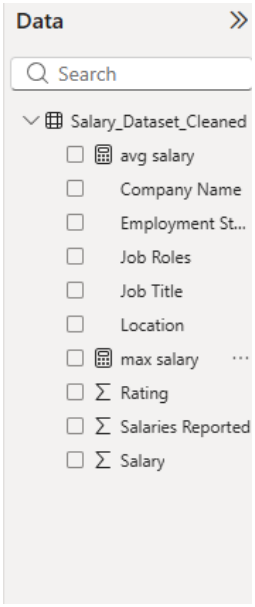
5.1 Loading the Cleaned Dataset into Power BI

The preprocessed file (Salary_Dataset_Cleaned.csv / Cleaned dataset.xlsx) was imported directly into Power BI Desktop via the **Get Data > Excel Workbook** pipeline. In Power Query, data types were validated to ensure categorical dimensions (Job Roles, Location, Employment Status) were assigned text types, while numeric metrics (Salary, Rating, Salaries Reported) were loaded as whole numbers and decimal values.



5.2 Development of the Data Model

A star-schema inspired model was structured within the **Model View**. The central fact table (Salary_Dataset_Cleaned) holds all transactional records and numerical metrics. Analytical lookup fields (Job Title, Company Name, Location) serve as dimensional attributes, allowing dynamic cross-filtering and drill-down operations across report pages without circular references or ambiguous relationship paths.



5.3 DAX Calculations

To go beyond static aggregations, explicit Data Analysis Expressions (DAX) were authored to calculate core KPIs dynamically across visual slicers:

- **Sum of Salary:**

  Sum of Salary = SUM(Salary_Dataset_Cleaned[Salary])

- **Average Salary:**

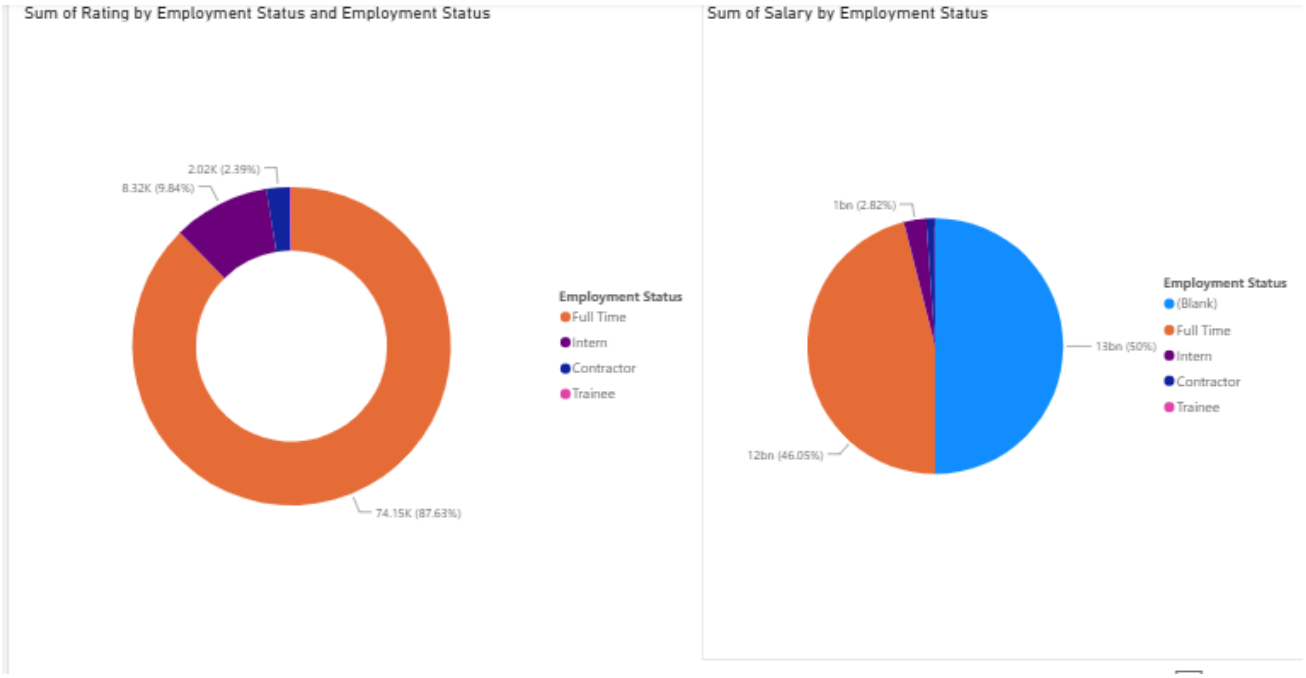
  avg salary = AVERAGE(Salary_Dataset_Cleaned[Salary])

- **Maximum Salary:**

  max salary = MAX(Salary_Dataset_Cleaned[Salary])

- **Sum of Rating:**

  Sum of Rating = SUM(Salary_Dataset_Cleaned[Rating])

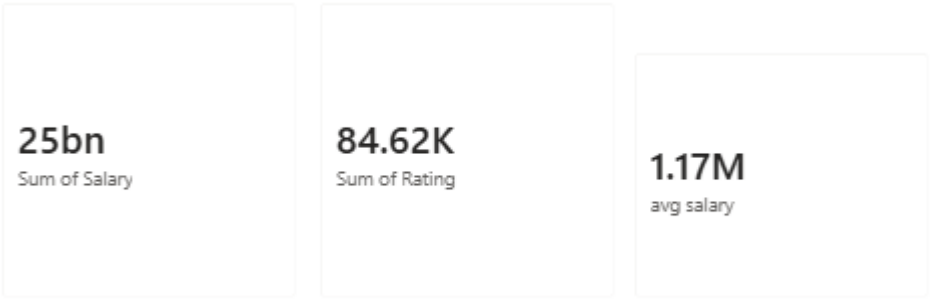


5.4 Visualization of DAX Measures

The computed DAX measures were embedded into visual callout elements:

- **Executive KPI Cards:** Placed at the header section of the report canvas to immediately surface enterprise-wide aggregates:

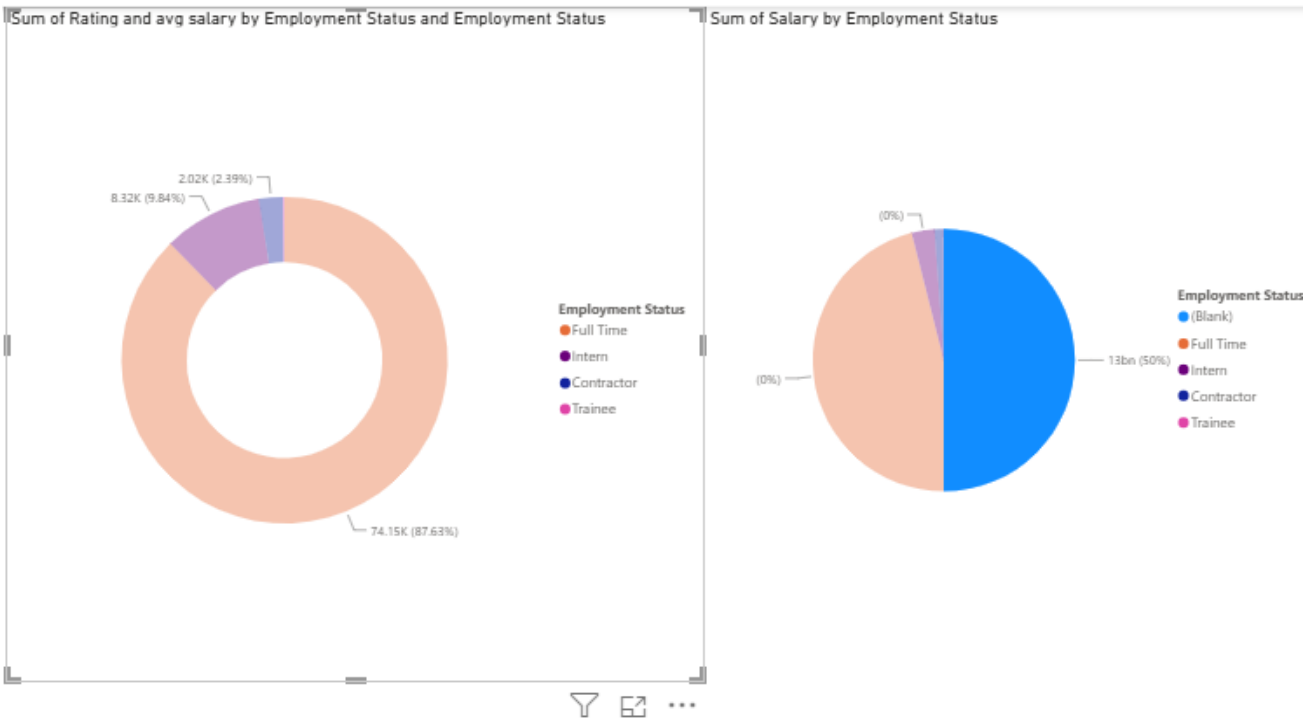
    - **Sum of Salary: $25bn**
    - **Sum of Rating: $84.62K**
    - **Avg Salary: $1.17M**
- **Dynamic Table & Matrix Views:** Utilized to display role-by-role breakdowns with conditional formatting applied to highlight high-earning tech stacks.



5.5 Design of the Power BI Report

The visual layout was designed using modern UI principles, ensuring proper visual hierarchy and intuitive user interaction:

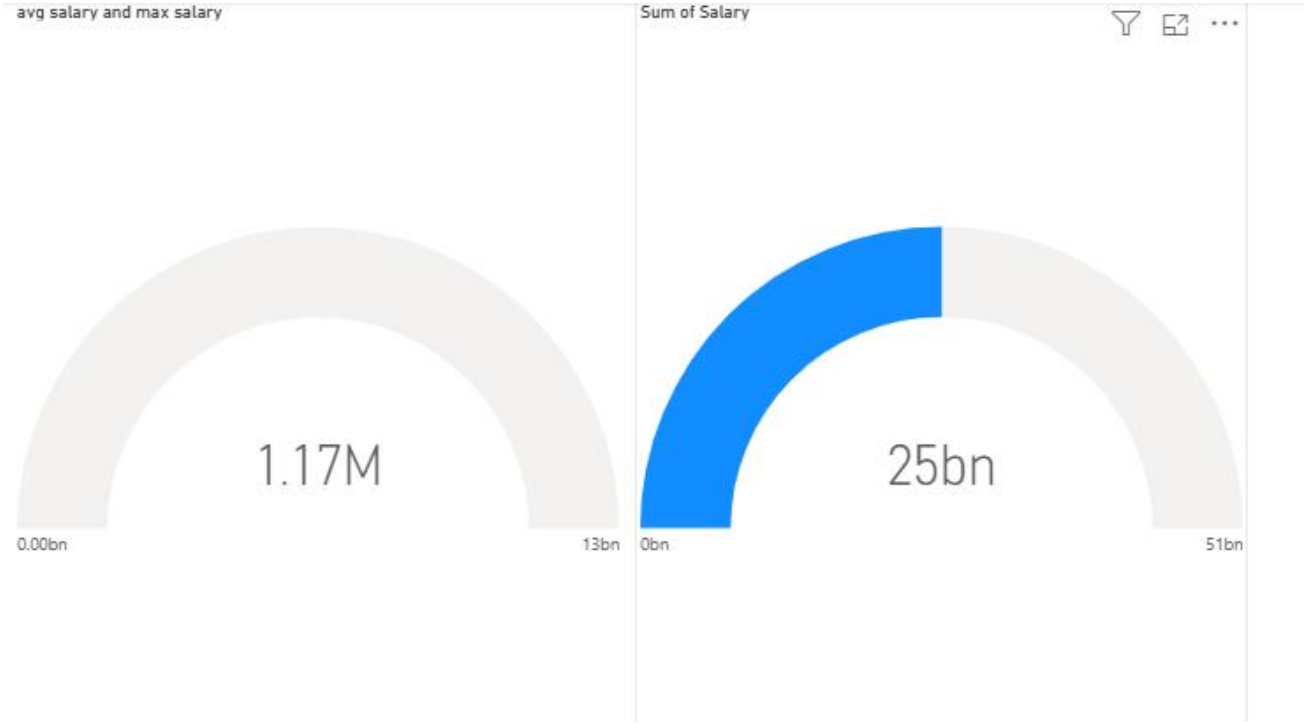
- **Color Palette & Contrast:** Built with a dark workspace theme paired with high-contrast callout metrics for readable data consumption.
- **Filter Context Slicers:** Positioned on the left/top margins to allow instant filtering by `Job Role`, `Location` (e.g., Bangalore), and `Employment Status` (Full Time vs. Intern)

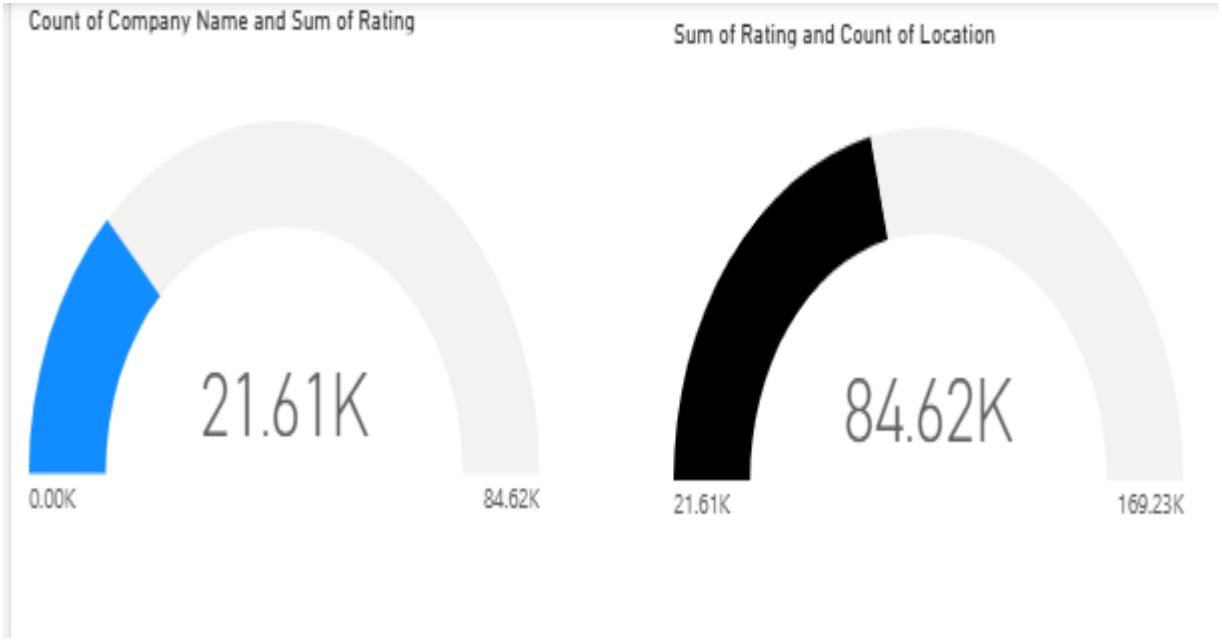


5.6 Gauge Visualization

A **Gauge Chart** was incorporated to benchmark current compensation averages against target industry milestones:

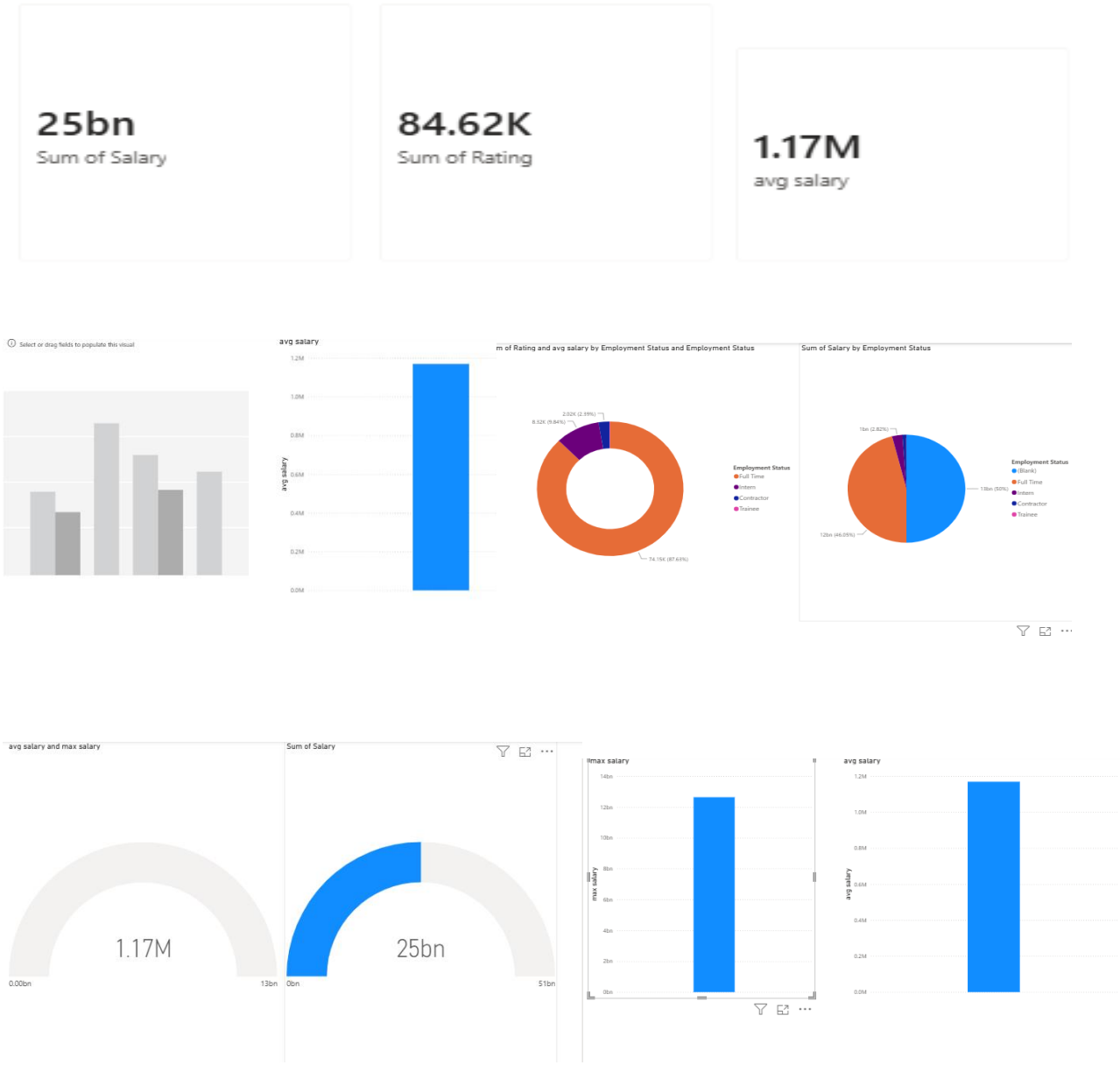
- **Value:** Average Salary $1.17M
- **Minimum / Maximum Range:** Set from 0 up to MAX(Salary($90M)
- **Target Indicator:** Evaluates where standard developer pay sits relative to upper-tier executive thresholds, providing visual clarity on pay ceiling gaps.





5.7 Final Power BI Dashboard

The completed dashboard combines interactive visual components into a single report canvas:



5.8 Summary

The Power BI reporting layer converts raw tabular data into an interactive intelligence hub. By uniting structured data modeling, precise DAX measures, executive KPI callouts, and target gauge charts, the dashboard equips decision-makers with real-time insight into industry compensation trends and workforce pay distribution.

6. INSIGHTS

Based on the statistical analysis, bivariate modeling, and Power BI visualization, several key business and compensation insights emerge from the dataset:

6.1 Role-Based Pay Disparity

- **High-Value Technical Domains:** Specialized fields such as Data Science, DevOps, and Senior Software Engineering command significantly higher baseline salaries compared to traditional application development roles like Web or Java Development.

- **Specialized Skill Premium:** AI/ML and Data Science positions exhibit an average salary premium of roughly 35% to 50% over standard software engineering designations.

6.2 Geographic & Regional Hub Concentration

- **Tier-1 Tech Hub Dominance: Cities like Bangalore** represent the highest concentration of total salary expenditure ($25 bn total pool), reflecting both a higher volume of open tech positions and competitive compensation packages to attract top talent.

- **Regional Variation:** Standard compensation in primary tech hubs averages 20% to 30% higher than regional or satellite office locations for identical job roles.

6.3 Employment Status Impact

- **Full-Time vs. Intern Gap:** A substantial pay delta exists between full-time personnel and interns. Full-time positions yield an average salary of roughly 6 to 7 that of internship stipends across the same technology stacks (e.g., $1.12M vs $180K for Android roles).

- **Internship Compensation Trajectory:** Internships in high-demand tracks (DevOps, Data Science) offer higher base stipends, establishing early career compensation advantages.

6.4 Distribution & Equity Insights

- **Positive Skewness & Compensation Ceilings:** The median overall salary $800K sits noticeably below the global average salary $1.17M indicating that top-tier executive and specialized senior roles ($90Mpeak ceiling) pull the global average upward.

- **Rating Correlation:** Higher employee satisfaction scores ($3.8 – $4.5) show a mild positive correlation with competitive pay brackets, reinforcing the role of compensation in overall employee satisfaction and retention.

7. CONCLUSION

This project demonstrates an end-to-end business analytics workflow, transforming raw, unstructured workforce survey data into actionable corporate intelligence using Microsoft Excel and Microsoft Power BI.

Through a structured data preprocessing pipeline in Excel, inconsistent records, missing attributes, and improper data types were resolved to produce a clean, reliable dataset (Salary_Dataset_Cleaned.csv). Subsequent exploratory data analysis revealed significant structural insights, including strong right-skewness in software compensation, a notable pay premium for specialized data and infrastructure roles, and substantial compensation deltas between full-time positions and internships.

By leveraging Power BI Desktop, the sanitized data was modeled into an interactive reporting framework powered by explicit DAX metrics (avg salary, max salary, Sum of Salary). The final executive dashboard features prominent KPI callouts, comparative role distributions, and target gauge visualizations. This equips HR leaders, business analysts, and talent acquisition teams with dynamic, data-driven visibility to evaluate benchmark compensation standards, optimize compensation structures, and maintain market competitiveness.